

LAB:-2

Q1.Containerization & Kubernetes Orchestration

Containerize a sample application using Docker: write a Dockerfile, build the image, and run the container. Document the image layers and container behavior.

```
ayushi2323@LAPTOP-6LVIDNH7$ mkdir docker-demo
ayushi2323@LAPTOP-6LVIDNH7$ cd docker-demo
ayushi2323@LAPTOP-6LVIDNH7/docker-demo$ touch app.py Dockerfile
ayushi2323@LAPTOP-6LVIDNH7/docker-demo$ nano app.py
ayushi2323@LAPTOP-6LVIDNH7/docker-demo$ |
```

```
GNU nano 8.7.1 app.py *
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello! This application is running inside Docker."

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

```
ayushi2323@LAPTOP-6LVIDNHT x + v
ayushi2323@LAPTOP-6LVIDNHT$ mkdir docker-demo
cd docker-demo
ayushi2323@LAPTOP-6LVIDNHT/docker-demo$ touch app.py Dockerfile
ayushi2323@LAPTOP-6LVIDNHT/docker-demo$ nano app.py
ayushi2323@LAPTOP-6LVIDNHT/docker-demo$ nano Dockerfile
ayushi2323@LAPTOP-6LVIDNHT/docker-demo$ |
```

```
ayushi2323@LAPTOP-6LVIDNHT x + v
GNU nano 8.7.1 Dockerfile *
FROM python:3.11-slim

WORKDIR /app

COPY app.py .

RUN pip install flask

EXPOSE 5000
|
CMD ["python", "app.py"]
```




Hello! This application is running inside Docker.

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker run -d --name docker-demo-container -p 5000:5000 docker-demo:1.0
45a04f4d0e37cbae9d77ac3e9b5d46bb3780cc6d5fc93b21cfe3d4b20130cd80
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
45a04f4d0e37   docker-demo:1.0 "python app.py"         41 seconds ago Up 38 seconds 0.0.0.0:5000->5000/tcp
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ curl http://localhost:5000
Hello! This application is running inside Docker.ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$
```

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker history docker-demo:1.0
IMAGE          CREATED        CREATED BY          SIZE    COMMENT
3b67be2e461d   5 hours ago   CMD ["python" "app.py"] 0B      buildkit.dockerf
<missing>      5 hours ago   EXPOSE [5000/tcp]      0B      buildkit.dockerf
<missing>      5 hours ago   RUN /bin/sh -c pip install flask # buildkit 19.7MB   buildkit.dockerf
<missing>      5 hours ago   CCOPY app.py . # buildkit 12.3KB   buildkit.dockerf
<missing>      5 hours ago   WORKDIR /app           8.19KB   buildkit.dockerf
<missing>      6 days ago   CMD ["python3"]        0B      buildkit.dockerf
<missing>      6 days ago   RUN /bin/sh -c set -eux; for src in idl3 p... 16.4KB   buildkit.dockerf
<missing>      6 days ago   RUN /bin/sh -c set -eux; savedAptMark=${a... 48.8MB   buildkit.dockerf
<missing>      6 days ago   ENV PYTHON_SHA266=91bcdebfdde239a003ae93738... 0B      buildkit.dockerf
<missing>      6 days ago   ENV PYTHON_VERSION=3.11.16 0B      buildkit.dockerf
<missing>      6 days ago   ENV GPG_KEY=A035C8C19219BA821ECEA86864E628F... 0B      buildkit.dockerf
<missing>      6 days ago   ENV /bin/sh -c set -eux; apt-get update; a... 13.2MB   buildkit.dockerf
<missing>      6 days ago   ENV LANG=C.UTF-8       0B      buildkit.dockerf
<missing>      6 days ago   ENV PATH=/usr/local/bin:/usr/local/sbin:/usr 0B      buildkit.dockerf
<missing>      2 weeks ago  # debian.sh --arch 'amd54' out/ 'trixie' '@... 87.5MB   debuerreotype 0.
```

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker stop docker-demo-container
docker-demo-container
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        NAMES
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
45a04f4d0e37   docker-demo:1.0 "python app.py"         5 hours ago    Exited (255) 5 minutes ago 0.0.0.0:5000->5000/tcp docker-demo-container
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker start docker-demo-container
docker-demo-container
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAME
45a04f4d0e37   docker-demo:1.0 "python app.py"         5 hours ago    Up 11 seconds 0.0.0.0:5000->5000/tcp, [:::5000->5000/tcp] docker-demo-container
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$
```

